

wed

SPM

15-oct

- PMI (Project Management Institute)
 - ↳ PMBOK (Project Management Book of Knowledge)
- MS Teams
- Scenario based

- Project

↳ start time - End time - with deliverables

↳ time - budget - scope

- Scope

↳ amount of work we are going to do

lecture - 1

Intro to SPM
pdf

- ① Requirement
- ② Planning
- ③ Design
- ④ Develop
- ⑤ Test
- ⑥ Deploy
- ⑦ maintain

→ what is project management?

Techniques

* Initiating:

- cost
- Time
- Requirements
- Resource

* Planning

- Design

* Executing

- Development

* Monitoring and Control

- Monitor cost, time, and scope
- Directly related to risk
- In control we use strategies like change mgmt

* closing

* Differ

→ Project

* closing

✓ - Acceptance Testing

* Differentiating Factors

→ Project Management Framework.

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Knowledge Areas

- 1) Integration
- 2) Scope
- 3) Schedule
- 4) Cost
- 5) Quality
- 6) Resource
- 7) Communication
- 8) Risk - negative impact on scope - times
cost
- 9) Procurement
- 10) Stakeholder

* off-the-shelf software

↳ General purpose

* bespoke software

↳ special purpose

* Process and systems reengineering services

↳ update and maintain.

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23 - October

Q: What are PM responsibilities? Etween
responsibilities

* Effective Project Manager

- Knowledge
- Performance
- Personal Skills

KPI → Key Performance Indicator.

* Program is made up of multiple Project.

* Portfolio is made up of multiple programs

* Project Selection methods

- (1) Benefit Measurement Methods
- (2)

Monday

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27 October

Tuesday

Lecture 2

* Cost benefit ratios

$$CBR = \frac{\text{Total Earning}}{\text{Total Cost}}$$

> 1 P

< 1 L

= 1 Break even

* Return on investment

* Payback Period

↳ When ~~will~~ will we get our investment back

* Discounted Cash Flow

$$FV = PV(1+i)^n$$

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30 oct

BAC

keyword

CV, SV

CPI, SPI

Schedule Variance

$$SV = \underbrace{EV}_{\text{earn value}} - \underbrace{PV}$$

Cost Variance

$$CV =$$

CPI

cost performance
index

SPI

schedule performance
index

CPI

$$CPI = \frac{EV}{AC}$$

$$CPI < 1$$

-ve $\rightarrow$ over budget

+ve $\rightarrow$ under budget

0 $\rightarrow$ on budget

SPI

$$SPI = \frac{EV}{PV}$$

SPI

-ve $\rightarrow$ behind
schedule

0 $\rightarrow$ on schedule

+ve $\rightarrow$ ahead
schedule

Budget at completion / Planned Budget

~~BAC~~?

BAC = Project cost = Sum of PV

Estimate to completion

(ETC)

$$\cancel{ETC} = \cancel{AC} \times \cancel{CPI} \times \cancel{(BAC - EV)}$$

$$ETC = \frac{BAC - EV}{CPI}$$

$$EAP = AC + ETC$$

↑
Estimate
at completion

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3/11/25

divide by

→ CPI

so to consider past performance

* if cost effects ETC than schedule also does

$$ETC = \frac{(BAC - EV)}{CPI \times SPI}$$

$$ETC = \frac{(B.A.C - E.V)}{(0.8 \times CPI)(0.2 \times SPI)}$$

↑ ↑
influence factors

* Variance at completion (VAC)

$$VAC = BAC - EAC$$

* Internal stakeholders

↳ Manager, ceo, etc

* External stakeholders

↳ Government
Client is not included.

* Prioritize Your Stakeholders

↳ How do you manage your stakeholders according to

↳ Power & Interest Grid

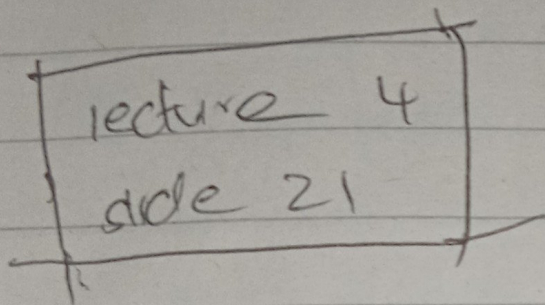
* Stakeholder register

↳ Is maintained so that we can contact concerned person in case of a problem

↳ The Power interest Grid tells which stakeholder to dealing

in what manner.

* Level of Engagement



lecture 4

* Change control

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* Project Scope Management

lecture 4

↳ Requirements

* Constraints

↳ Time ↳ Scope ↳ Cost

* Scope Creep

↳ Technical Debt

leave slide
7

* Product Scope v/s Project Scope

↓
characteristics of
finished item

↓
Time, Cost and
requirements of
the product

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12/11/25

- * Collecting requirements

- * Functional & non functional req

- * Tools & Techniques

- ↳ Interviews

- * Domain Experts

- ↳ Focus Group

- * Multiple Experts

- ↳ Historical Records.

- less imp * Similar Projects ~~or~~

- more imp * Old way of doing the same thing. maybe manual handwritten things.

- ↳ Facilitated Workshops

- * Stakeholders meeting

* Group Decision Making Techniques

↳ Unanimity

* all agree on one.

↳ Majority

* 50% or more

↳ Plurality

* highest percentage

↳ Dictatorship

* Highest power wins

* Functional - non functional

* Requirements Document.

* SRS

Jetbrains

* Requirement Traceability Matrix

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17/11/2

* Software Project scheduling

↳ Work-breakdown structure.

①

②

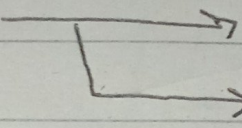
③

~~1/8~~ ~~2/8~~ ~~3~~ ~~4~~ ~~5~~ ~~6~~ ~~7~~ ~~8~~

20/11/25

spm

Thursday

Tasks have  ^(priority) * Precedence 5
* Dependence

* Types of dependencies

* Sequence Activities Process

* Project sequence network diagram^a

↳ Table

↳ Diagram

↳ Critical Path

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6PM

24/11/25

* Estimate Activity Duration Process

Lecture 5

↳ The time of project completion

* Techniques

↳ Expert Estimation:

An expert within org or domain expert

↳ Analogous:

Comparison with similar old project

↳ Parametric:

use of data like efficiency resources, etc

↳ Project Evaluation & Review Technique (PERT) Estimation:

Give 3 values optimistic
most likely
pessimistic ↑ and take average

→ Reserve Analysis:
Adding a reserve time in
the task.

* PERT

→ Triangular Distribution

$$E = (O + M + P) / 3$$

→ Beta distribution

$$E = (O + 4M + P) / 6$$

→ A.K.A Three point numerical.

skip develop schedule Process

* Control schedule Process